Fusion of Modalities and Evaluation of Multimodal Retrieval-Augmented Generation

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***Abstract*—Multimodal Retrieval-Augmented Generation (M-RAG) systems improve the ability of large language models to answer questions using both textual and visual information. However, the performance of these systems depends strongly on how different modalities are combined. In this work, we design and compare three M-RAG pipelines based on Early Fusion, Late Fusion, and Hybrid Fusion. The pipelines use a common document extraction, chunking, embedding, retrieval, re-ranking, and response-generation process, while differing in their fusion strategies. We evaluate retrieval performance using Hit@5, Mean Reciprocal Rank (MRR), and a proposed metric called Error Point, which considers both the rank and vector distance of incorrectly retrieved chunks. We also introduce the Chunk-Weighted Generation Fidelity framework to evaluate answer generation through chunk ablation. The results show that Late Fusion provides the best overall retrieval performance, achieving an MRR of 0.8291 and the lowest Mean Error Point of 0.1868. Early Fusion achieves the highest Hit@5 score of 0.9000 but is evaluated on a smaller number of questions. These findings show that the choice of fusion strategy has a clear effect on M-RAG performance and should be selected according to the dataset structure and modalities.**

***Index Terms*—Multimodal Retrieval-Augmented Generation, Large Language Models, Early Fusion, Late Fusion, Hybrid Fusion, Information Retrieval, Error Point, Chunk Ablation, Multimodal Question Answering.**

I. Introduction

Large Language Models (LLMs) are trained on large amounts of data available over the internet. However, LLMs often generate incorrect responses when asked deep, domain-specific questions. This is known as hallucination [1]. LLMs hallucinate due to a lack of context, and Retrieval Augmented Generation (RAG) is used to overcome this problem [1], [4]. In most cases, RAG works well with textual data, but when the dataset contains multiple modalities such as images, charts, audio, and video, even RAG is not sufficient to provide accurate context to LLMs [2], [5], [11]. Thus, the Multimodal Retrieval Augmented Generation (M-RAG) technique is used, which handles both textual and non-textual data and provide accurate context to LLMs [5]-[16].

Multimodal Retrieval Augmented Generation (M-RAG) can be designed in three different ways, depending on when the different modalities are fused in the pipeline: A. Early Fusion, where data is fused initially and then encoded into embeddings. B. Late Fusion, where different modalities are encoded into embeddings separately and then fused. C. Hybrid Fusion, which is a combination of Early Fusion and Late Fusion [17]-[22].

Every dataset is different - different in structure, different in meaning, different in modalities, and so on. Hence, confirming that any systematically designed RAG pipeline is x% accurate is dataset-specific, and the accuracy will most likely change if the dataset changes [3], [4], [11], [14]. Therefore it is necessary that a dataset be tested across different pipelines, so that the pipeline design best suited to that dataset can be selected. Which pipeline best suits a given dataset? This question can be answered based on how the results of the RAG pipeline are evaluated, and on what parameters [1], [2], [4]. The performance of a RAG pipeline is calibrated in two phases: A. the Retrieval Phase, and B. the Generation Phase [1], [2], [4]. The parameters used for evaluation should be measure across most of these aspects. The black-box problem associated with AI raises the question of to what extent it is appropriate to rely entirely on the LLM-as-judge approach for evaluating RAG responses [1], [11], [14].

Accuracy of RAG-based AI applications may degrade when data consists of different modalities [2], [6], [9], [17]-[22]. The degradation in performance depends on the methods used for fusing different modalities in the pipeline. Most RAG architectures are primarily designed for textual data, thus modifying the RAG pipeline to handle and combine multiple modalities usually may degrade the performance of RAG-based AI applications [5], [6], [7], [8], [11], [12], [13].

We analyzed existing Multimodal Retrieval-Augmented Generation architectures [5]-[16] and, based on their strengths and limitations, developed three M-RAG pipelines using Early Fusion, Late Fusion, and Hybrid Fusion [17]-[22]. The pipelines share the same document-processing, retrieval, re-ranking, and generation components [4], [8], [11], which allows the effect of the fusion strategy to be examined under a common experimental setup.

We also propose Error Point which is a rank-aware and distance-sensitive retrieval metric. Unlike MRR and Hit@K, which primarily consider whether the correct chunk is retrieved and its position in the ranking [1], [4], [14], Error Point additionally considers the vector distances and rank positions of incorrectly retrieved chunks and applies an explicit penalty

when the correct chunk is not retrieved. A lower Error Point score indicates better retrieval quality.

For the generation phase, we introduce the Chunk-Weighted Generation Fidelity (CWGF) framework. CWGF generates and evaluates answers using progressively larger subsets of the retrieved context, from the top-1 to the top-5 chunks, to construct a chunk ablation curve. By observing how answer quality changes as additional chunks are introduced, the framework helps to distinguish retrieval failures from generation-level failures and reduces dependence on a standalone LLM-as-judge evaluation [1], [2], [11], [14].

Our results show that when a multimodal dataset is passed through three different M-RAG pipelines implemented with different fusion techniques (Early, Late, and Hybrid Fusion), the results differ considerably. The most balanced results are displayed by the Late Fusion pipeline, which outperforms both Early Fusion and Hybrid Fusion. We believe these results will vary for different datasets, depending on their type, modalities, and structure [3], [11], [14]. This approach helps in identifying the most suitable M-RAG pipeline for a given dataset by comparing the performance of each pipeline.

II. Related Work

Existing research on Retrieval-Augmented Generation have mainly focused on improving retrieval techniques, similarity measures, multimodal architectures, and evaluation frameworks[1]-[4]. Mulakala et al. evaluated reasoning-based language models in RAG systems using metrics like MRR, Faithfulness, and Semantic Answer Similarity, showing that reasoning models generated answers that were more closely aligned with retrieved evidence [1]. Kartiyanta et al. compared different retrieval methods, including BM25, dense embeddings, and hybrid approaches, for multi-document RAG systems. They found that traditional and hybrid retrieval methods could achieve similar results while requiring fewer computational resources compared to advanced embedding-based methods [3]. Similarly, Elkiran et al. compared Cosine Similarity, Inner Product, and L2 Distance using retrieval, latency, token-usage, and cost metrics, finding that Cosine Similarity and Inner Product performed more effectively for document retrieval [4]. These studies demonstrate the importance of retrieval configuration, but their evaluations mainly depends on traditional metrics such as MRR, Recall, Precision, MAP, and NDCG [1], [3], [4].

Several studies have developed M-RAG systems for different application domains and document types [5]-[8]. Patel et al. proposed a system that processes text, images, video, audio, and scanned documents using LLaMA3, OCR, ImageBind, and Milvus [5]. Thiyagarajan applied M-RAG to retail data involving structured information, text, images, and customer interactions [6], while Kránics et al. developed a context-aware chatbot capable of answering questions from textual and visual information [7]. Loo et al. focused on multimodal PDF retrieval through configurable chunking, semantic and keyword search, and image captions [8]. These studies show the effectiveness of M-RAG systems in handling different types of data; however, their evaluations mainly focus on overall system accuracy, relevance of retrieved information, and application-level performance rather than deeper analysis of retrieval quality [5]-[8].

Other work has focused on improving document understanding, retrieval, and reasoning [9], [10], [12], [13]. Meng combined document-aware processing, dense and sparse retrieval, reranking, and text-image alignment to reduce hallucination in complex PDF question answering [9]. Wang et al. proposed AdaMKG, which combines multimodal knowledge graphs, hybrid retrieval, and multi-hop reasoning for enterprise information management [10]. Liu et al. introduced HM-RAG, a hierarchical multi-agent framework that decomposes complex queries and retrieves information from vector databases, knowledge graphs, and the web [12]. Wu et al. proposed MARA, which creates query-adaptive document representations and dynamically selects additional evidence from documents containing text, tables, figures, and complex layouts [13]. While these methods improve retrieval and reasoning performance, most of them focus on evaluating individual architectures and do not provide a direct comparison of different modality-fusion approaches under the same experimental setup [9], [10], [12], [13].

A different group of studies has introduced benchmarks and evaluation frameworks for multimodal retrieval and generation [2], [11], [14]. Sun et al. proposed MRAFE, which evaluates properties such as noise robustness, negative rejection, information integration, and counterfactual robustness [2]. Liu et al. introduced M2RAG and MM-RAIT for tasks including multimodal question answering, image captioning, fact verification, and image reranking [11]. Yu et al. proposed MRAMG-Bench, which uses human-annotated datasets and evaluates both textual and visual retrieval and generation [14]. Task-specific studies include mRAG-gim for lightweight image captioning [15] and the AdaVideoRAG and REViG frameworks for video understanding and generation [16]. These studies use metrics such as Accuracy, F1-score, ROUGE-L, BERTScore, CIDEr, Context Recall, and Visual Recall, providing broad measures of task-level performance [11], [14]-[16].

Overall, previous studies show that M-RAG systems can achieve better performance by using techniques such as multimodal alignment, hybrid retrieval, adaptive document representation, knowledge graphs, and multi-agent reasoning [5]-[13]. However, most existing works either focus on a single system design or evaluate performance mainly through traditional retrieval and generation metrics [1]-[4], [11], [14]. There is limited research that directly compares different fusion strategies, such as Early Fusion, Late Fusion, and Hybrid Fusion, using the same dataset and processing setup [17]-[22]. Additionally, traditional retrieval metrics mainly evaluate whether the correct information is retrieved and its ranking position, but they do not consider the impact of incorrectly retrieved chunks and their ranking positions [1], [4]. To address these limitations, this study compares three different fusion-based M-RAG pipelines, introduces Error Point as an additional retrieval evaluation metric, and uses the Chunk-

Weighted Generation Fidelity framework to provide a deeper analysis of generation quality.

III. Methodology

A. Overview

We propose three Multimodal RAG pipelines that share a common data preparation and retrieval procedure, but differ in how and when textual and visual information is fused before indexing [8], [9], [11]. The three variants, namely *Early Fusion*, *Late Fusion*, and *Hybrid Fusion*, are described in Sections III-C, III-D, and III-E, respectively. Figures 1, 2, and 3 show the indexing-phase architecture of each pipeline.

Before describing the pipelines, we define the notation used throughout this section. Let

$$T = \{t_1, t_2, \ldots, t_n\} \tag{1}$$

denote the set of text chunks extracted from the document corpus, and let

$$I = \{i_1, i_2, \ldots, i_m\} \tag{2}$$

denote the set of image descriptions generated for each extracted image. For any pair (t, i), where $t \in T$ and $i \in I$, let

$$\mathrm{sim}(t, i) \tag{3}$$

denote the cosine similarity between their embeddings. A user query is denoted by q, and the final set of documents retrieved and passed to the LLM for a given query is denoted by D_q.

B. Common Setup for All Pipelines

Prior work suggests that the performance of a RAG pipeline depends heavily on how the source documents are extracted and chunked [4], [8], [9]. We use Docling to extract content from the dataset, since it preserves the structural layout of the source document during extraction. The extracted text is split using a hierarchical chunking strategy, which retains contextual relationships better than flat chunking [8], [9].

Images embedded in the documents are extracted using a YOLOv8 model, together with a window of nearby text for context. A vision-language model is then used to generate a short description for each extracted image [5], [15], [16]. Every text chunk and image description is assigned a unique SHA-256 identifier, which is used to store and later retrieve its actual content from a key-value store.

At query time, the user query q is embedded and the top-k candidate documents are retrieved from the vector database [3], [4]. These candidates are passed through a re-ranker to obtain the final filtered set D_q [4]. The identifiers of the documents in D_q are then used to fetch their content from the key-value store. This content, together with the query, is passed to the LLM to generate the final response.

This retrieval procedure is identical across all three pipelines described below. The pipelines differ only in how content is fused before indexing.

C. Early Fusion Pipeline

1) Indexing Phase: For every text-chunk–image-description pair $(t, i) \in T \times I$, we compute

$$\mathrm{sim}(t, i). \tag{4}$$

If this similarity exceeds a fixed threshold τ, the pair is concatenated into a single fused unit before embedding [17]:

$$f_{t,i} = [t; i], \quad \text{if } \mathrm{sim}(t, i) > \tau. \tag{5}$$

A single text chunk or image description is allowed to be reused across multiple pairs if it crosses the threshold with more than one counterpart on the other side. Therefore, this is not a strict one-to-one matching. Any text chunk or image description that does not exceed τ with any counterpart is embedded and stored as a standalone unit. All fused units and standalone units are assigned unique IDs, stored in the key-value store, and embedded into the vector database for indexing.

2) Retrieval Phase: The retrieval phase is identical to the common setup described above.

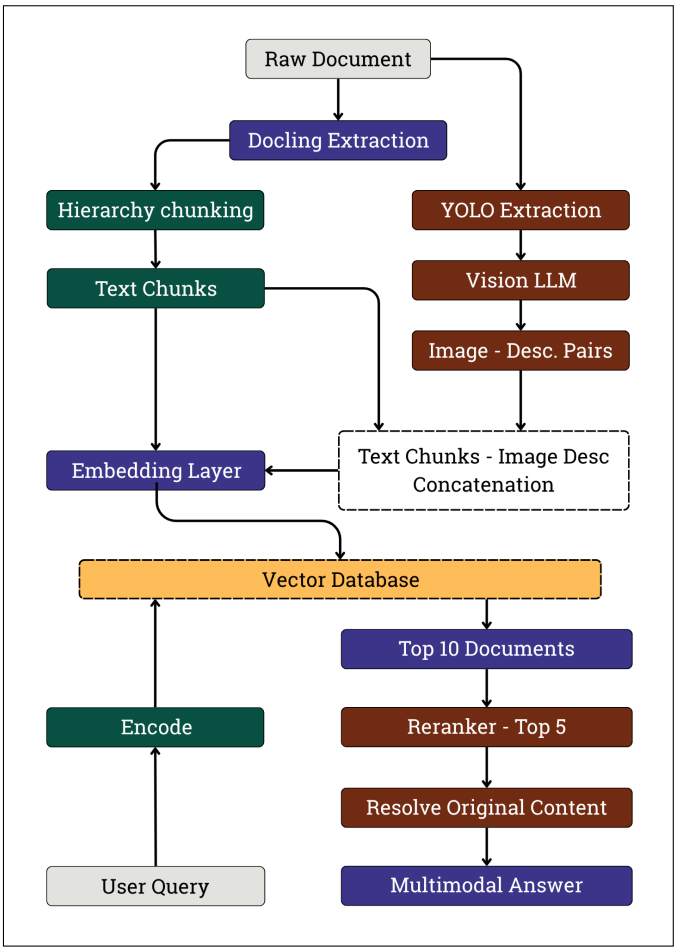


Fig. 1. Early Fusion Pipeline

D. Late Fusion Pipeline

1) Indexing Phase: Text chunks and images are first embedded independently and stored in two separate vector database collections: *text-only* and *image-only* [17], [18]. For every image description $i \in I$, we compute its similarity against every text chunk in T and select the top two most similar chunks:

$$\{t_{i,1}, t_{i,2}\} = \text{Top2}_{t \in T} \, \text{sim}(t, i). \tag{6}$$

Two separate fused embeddings are then created for that image: one from i and $t_{i,1}$, and another from i and $t_{i,2}$. These are produced by averaging their respective embeddings:

$$\mathbf{z}_{i,j} = \frac{1}{2} \left(\mathbf{e}_i + \mathbf{e}_{t_{i,j}} \right), \quad j \in \{1, 2\}, \tag{7}$$

where $\mathbf{e}_i$ denotes the embedding of image description i, and $\mathbf{e}_{t_{i,j}}$ denotes the embedding of text chunk $t_{i,j}$.

Since averaging two embeddings is only meaningful if they lie in the same vector space, both the text and image embeddings used here are generated using the CLIP embedding model, which places them in a shared space by construction [5], [15]. Both fused embeddings are stored in a third collection, called *fused-only*.

We deliberately keep all three collections rather than only the fused one. If a relevant piece of content is not well represented in *fused-only*—for instance, if the pairing missed a genuinely relevant chunk-image association—it can still be retrieved directly from *text-only* or *image-only*. To avoid a matched item being over-represented simply because it appears in more than one collection, after the re-ranking step we collapse the retrieved set down to its unique document IDs before passing the content to the LLM [8], [13].

2) Retrieval Phase: The retrieval phase is identical to the common setup described above, with candidates drawn from all three collections before re-ranking and deduplication.

E. Hybrid Fusion Pipeline

1) Indexing Phase: As in the late fusion pipeline, text chunks and images are first embedded and stored separately in *text-only* and *image-only* collections. To decide which content should be fused early versus late, we build a graph

$$G = (V, E, W), \tag{8}$$

where

$$V = T \cup I \tag{9}$$

is the set of nodes, containing every text chunk and every image description.

The edge set is defined as

$$E = \{(v_a, v_b) \mid v_a, v_b \in V, \; v_a \neq v_b\}, \tag{10}$$

meaning that every pair of distinct nodes is connected by an edge. Each edge is weighted by the cosine similarity between its two nodes [20], [21]:

$$W(v_a, v_b) = \text{sim}(v_a, v_b). \tag{11}$$

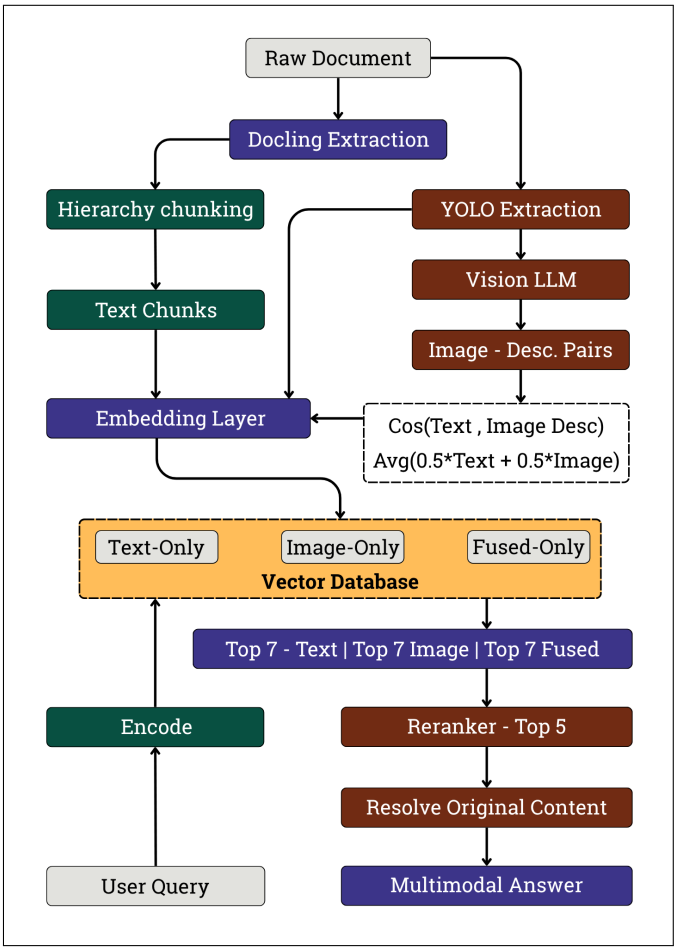


Fig. 2. Late Fusion Pipeline

We apply the Louvain community detection algorithm on G to partition V into clusters [10], [20]. Nodes that fall within the same cluster are treated as candidates for early fusion and are fused using the same threshold-based procedure described in Section III-C. Nodes that fall across different clusters are treated as candidates for late fusion and are fused using the top-2 matching-and-averaging procedure described in Section III-D [17], [18], [22].

The resulting early-fused and late-fused embeddings are stored in two separate collections, called *early-fusion* and *late-fusion*, respectively.

2) Retrieval Phase: The retrieval phase is identical to the common setup described above, with candidates drawn from both collections before re-ranking.

F. Multimodal Response Composition

The response generated by each of the three pipelines is purely textual by default, even though the context used to generate it often includes images. To make the final output genuinely multimodal, we insert the image or images used during generation back into the answer at an appropriate position [15], [16].

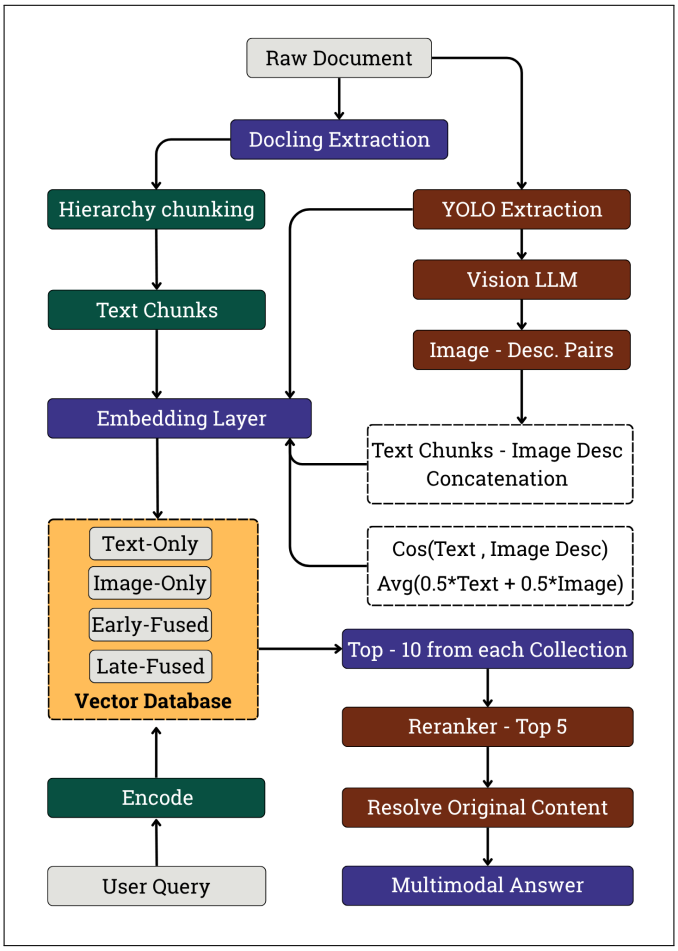


Fig. 3. Hybrid Fusion Pipeline

We construct a bipartite graph

$$G_b = (S \cup I_q, E_b), \tag{12}$$

where S denotes the set of sentences in the generated answer, and I_q denotes the set of images used during generation. Each edge between a sentence $s \in S$ and an image $i \in I_q$ is weighted by their relevance:

$$W_b(s, i) = \text{rel}(s, i). \tag{13}$$

We then apply Edmonds' Blossom algorithm to obtain a maximum matching over this graph. The resulting matching determines the most appropriate position within the generated answer to place each image [15], [16].

IV. Evaluation

A. Dataset Creation

The source corpus used in this study consists of research paper PDFs collected for the literature review process. Instead of using an existing multimodal benchmark dataset, these documents were selected as a domain-specific corpus to evaluate the three proposed M-RAG pipelines under identical experimental conditions [2], [11], [14]. The collected documents include different types of information, such as textual content, images, figures, graphs, charts, and tables, allowing the evaluation of multimodal retrieval and generation performance across various data formats [5], [8], [15], [16].

During the ingestion phase, the text content of each document was split into smaller sections called chunks. From these chunks, 40 relevant chunks were selected and individually provided to a Large Language Model (LLM) [1]. The LLM was instructed to generate one meaningful question along with its corresponding reference answer using only the information available in the given chunk [1], [2]. Chunks that contained very limited information or were too short to generate meaningful questions and answers were removed from the evaluation process.

The final evaluation dataset consisted of 40 question-answer pairs, containing four main fields: Question, Answer, Chunk, and Chunk ID. The Chunk field stores the original source content used for generating the question and answer, while the Chunk ID provides a unique identifier for tracking the corresponding document chunk. During the evaluation process, the questions from this dataset were provided as input to each Multimodal Retrieval-Augmented Generation (M-RAG) pipeline. The generated responses were then compared with the reference answers available in the dataset [1], [4]. The performance of each pipeline was evaluated based on the accuracy and similarity of the generated answers compared to the expected responses.

B. Parameters Used for Evaluation of the Retrieval Phase

Three primary metrics were used to evaluate the retrieval performance of the proposed MRAG pipelines [4].

1) Mean Reciprocal Rank (MRR): The first evaluation metric is *Mean Reciprocal Rank (MRR)* [4]. Reciprocal Rank (RR) is defined as the inverse of the rank of the first relevant document retrieved for a query. It measures how quickly a retrieval system returns a relevant result. Mean Reciprocal Rank extends this concept by averaging the reciprocal ranks over all queries, thereby providing an overall measure of retrieval effectiveness.

The MRR is calculated as:

$$\text{MRR} = \frac{1}{Q} \sum_{i=1}^{Q} \frac{1}{\text{rank}_i} \tag{14}$$

where:

- Q is the total number of queries.
- rank_i is the position of the first relevant document retrieved for the ith query.

A higher MRR value indicates that relevant documents are retrieved at higher ranks, reflecting better retrieval performance [4].

2) Hit@K: The second evaluation metric is *Hit@K* [4]. Hit@K measures whether at least one relevant document is present among the top K retrieved results for a given query. Unlike MRR, Hit@K does not consider the exact rank of the relevant document; it only checks whether a relevant document appears within the top K retrieved documents.

The Hit@K score is calculated as:

$$\text{Hit@K} = \frac{1}{Q} \sum_{i=1}^{Q} \mathbb{I}\left(\exists\, d \in \text{TopK}_i \text{ such that } d \text{ is relevant}\right) \tag{15}$$

where:

- Q is the total number of queries.
- TopK_i denotes the set of the top K retrieved documents for the ith query.
- $\mathbb{I}(\cdot)$ is the indicator function, which is equal to 1 if at least one relevant document is found within the top K results and 0 otherwise.

A higher Hit@K score indicates that the retrieval system is more successful at returning relevant documents within the top retrieved results [4].

C. Error Point (EP)

To address the limitations of existing retrieval evaluation metrics, we introduce *Error Point (EP)*, a novel rank-aware and distance-sensitive metric designed specifically for evaluating retrieval quality in Retrieval-Augmented Generation (RAG) pipelines [4]. Standard metrics such as Mean Reciprocal Rank (MRR) and Hit@K effectively measure whether the correct chunk was retrieved and at what rank [4]. However, they treat all incorrectly retrieved chunks as equally irrelevant and do not consider how close or far these incorrect chunks are from the query in the vector space [4].

MRR and Hit@K metrics mainly measure whether the correct chunk is retrieved and its position in the ranking. However, they do not consider the quality of the other retrieved chunks. In RAG systems, incorrectly retrieved chunks with high rankings can negatively affect the generated response, especially when they have low semantic similarity with the query [1], [4]. To address this limitation, Error Point (EP) considers multiple factors, including the rank of the correct chunk, the rank-based impact of incorrect chunks, and a penalty when the correct chunk is not retrieved. A lower Error Point value represents better retrieval performance, as it indicates fewer retrieval errors and more relevant context being provided to the generation model.

Error Point is defined under two cases.

a) Case 1: Correct Chunk Retrieved: Let c_{real} denote the correct chunk. If c_{real} is present within the top N retrieved chunks at rank k, Error Point is computed as

$$EP = \frac{k}{N} \cdot \frac{\sum_{i \neq k} \frac{1}{i} d_i}{\sum_{i \neq k} \frac{1}{i}}, \tag{16}$$

where d_i denotes the vector distance of the ith retrieved chunk from the query. The factor $\frac{k}{N}$ rewards the retrieval system for retrieving the correct chunk at higher ranks, while the weighted average penalizes incorrect chunks according to both their rank positions and distances.

b) Case 2: Correct Chunk Not Retrieved: If c_{real} is absent from the top N retrieved chunks, all retrieved chunks are considered incorrect. In this case, Error Point is defined as

$$EP = \frac{\sum_{i=1}^{N} \frac{1}{i} d_i}{\sum_{i=1}^{N} \frac{1}{i}} + \lambda d_{\text{real}}, \tag{17}$$

where d_{real} denotes the vector distance between the query and the correct chunk, and λ is a penalty multiplier that reflects the severity of a complete retrieval failure. In this work, λ is empirically set to 1.5.

The factor k/N penalizes the correct chunk when it appears at a lower rank, while 1/i gives greater importance to incorrect chunks appearing near the top of the retrieval list. The parameter λ controls the additional penalty for complete retrieval failure and was empirically set to 1.5.

The normalization term,

$$\sum_{i=1}^{N} \frac{1}{i}, \tag{18}$$

converts the rank-weighted distances into a weighted average, ensuring that Error Point remains comparable across queries with different values of N. The rank weight $\frac{1}{i}$ assigns greater importance to higher-ranked chunks, reflecting the practical behaviour of RAG systems in which top-ranked chunks exert a disproportionate influence on the generated response.

To the best of our knowledge, Error Point is the first retrieval metric that jointly considers rank position, vector distance of incorrectly retrieved chunks, and an explicit penalty for complete retrieval failure, thereby providing a more comprehensive and sensitive assessment of retrieval quality than existing evaluation metrics.

D. Generation Phase Evaluation

To evaluate the generation phase of the proposed Multimodal Retrieval-Augmented Generation (MRAG) pipelines, we introduce the *Chunk-Weighted Generation Fidelity (CWGF)* framework, a novel evaluation methodology designed to address the limitations of existing generation evaluation approaches [1], [2], [11].

Current evaluation methods for RAG generation, such as standalone BLEU or ROUGE scoring, treat generation quality as a single static measurement and fail to capture how the quantity and relevance of retrieved context influence the quality of the generated answer [1], [4]. Furthermore, existing methods do not distinguish whether a poor generated answer is caused by weak retrieval or by limitations of the language model itself [1]. Consequently, identifying the actual source of failure within the pipeline becomes difficult.

The CWGF framework addresses these limitations through a chunk ablation curve approach. For a given question q with

ground truth answer A_{actual}, a single fixed language model generates five answers,

$$\{A_1, A_2, A_3, A_4, A_5\}, \tag{19}$$

where each answer A_k is generated using only the top k retrieved chunks, with

$$k \in \{1, 2, 3, 4, 5\}. \tag{20}$$

Each generated answer is evaluated using the following composite scoring function [1], [4]:

$$\begin{aligned} S(A_k) = \frac{1}{3}\Big(&\mathrm{CosSim}(A_k, A_{\text{actual}}) \\ &+ \mathrm{BERTScore}(A_k, A_{\text{actual}}) \\ &+ \frac{\text{LLM-Judge}(A_k, A_{\text{actual}})}{10}\Big) \end{aligned} \tag{21}$$

Here, CosSim measures semantic similarity between answer embeddings, $\mathrm{BERTScore}$ captures token-level contextual similarity using BERT representations, and $\mathrm{LLM\text{-}Judge}$ is a prompted large language model that evaluates the generated answer in terms of correctness and faithfulness on a scale from 0 to 10 [1], [4]. The LLM-Judge score is subsequently normalized to the range $[0, 1]$ [1].

The resulting scores,

$$\{S(A_1), S(A_2), S(A_3), S(A_4), S(A_5)\}, \tag{22}$$

form the chunk ablation curve. The actual answer generated by the RAG pipeline, denoted by A_{Gen}, is also evaluated using the same scoring function to obtain $S(A_{\text{Gen}})$ and is plotted as a reference point on the curve.

The relative position of $S(A_{\text{Gen}})$ on the ablation curve provides direct diagnostic insight [1]. If $S(A_{\text{Gen}})$ lies near $S(A_5)$, the pipeline is effectively utilizing all retrieved context. If $S(A_{\text{Gen}})$ falls below $S(A_1)$, a generation-level failure is indicated regardless of retrieval quality. Furthermore, if the ground truth chunks are absent from the top-5 retrieved set, a retrieval-level failure is flagged independently.

By isolating the chunk variable while keeping the language model fixed, the CWGF framework separates retrieval failures from generation failures, a diagnostic capability absent in existing RAG generation evaluation methods [1], [4]. Consequently, CWGF provides a principled, interpretable, and reproducible framework for evaluating generation quality in Retrieval-Augmented Generation systems.

V. Results

A. Retrieval Performance

Table I shows the retrieval performance comparison of the three M-RAG pipelines. Among the three approaches, Late Fusion achieved the best results, with the highest Mean Reciprocal Rank (MRR) of 0.8291 and the lowest Mean Error Point (EP) of 0.1868. This shows that Late Fusion was more effective at ranking the correct source chunk higher while reducing the impact of irrelevant or incorrect retrieved chunks [4].

Early Fusion achieved the highest Hit@5 score of 0.9000, which means that the correct chunk was retrieved within the top five results for 90% of the evaluated questions [4]. However, its Mean Reciprocal Rank (MRR) score of 0.7333 was lower compared to Late Fusion. This indicates that while Early Fusion was successful in retrieving the correct chunk among the top results, it did not consistently rank the correct chunk in the highest positions [4]. In other words, Early Fusion could find the relevant information, but it was less effective at placing it at the top of the retrieval list.

Hybrid Fusion produced the lowest Hit@5 score of 0.7436 and the highest Mean Error Point of 0.2623. Therefore, among the three pipelines, Hybrid Fusion showed the weakest retrieval performance for the evaluated dataset.

TABLE I
Retrieval performance of the proposed M-RAG pipelines

Pipeline	Hit@5	MRR	Mean EP
Early Fusion	0.9000	0.7333	0.1975
Late Fusion	0.8462	0.8291	0.1868
Hybrid Fusion	0.7436	0.7308	0.2623

Note: Higher values of Hit@5 and MRR indicate better retrieval performance, whereas a lower Error Point (EP) value indicates better performance.

B. Generation Performance

Generation performance was evaluated using chunk-ablation curves. In this process, each pipeline generated answers using an increasing number of retrieved chunks, starting from only the top-ranked chunk up to the top five retrieved chunks. The generated answer from each pipeline, represented as A-Gen, was then evaluated using the same composite scoring method to measure how effectively the retrieved context contributed to the final response [1], [4].

Early Fusion showed the greatest variation across the ablation curve. Its score increased from approximately 0.60 with one chunk to approximately 0.79 with five chunks. This indicates that the Early Fusion pipeline depended strongly on receiving a larger amount of retrieved context. The A-Gen score of 0.7890 was close to the five-chunk result which suggest that the final pipeline response effectively used the full retrieved context.

Late Fusion displayed a comparatively stable ablation curve, with scores remaining approximately between 0.75 and 0.77. This suggests that its answer-generation performance was relatively insensitive to the number of chunks provided. Its A-Gen score of 0.7768 was slightly higher than most points on the ablation curve, indicating consistent use of the retrieved evidence.

Hybrid Fusion achieved the highest A-Gen score of 0.8034. Its ablation scores also remained consistently close to 0.80.

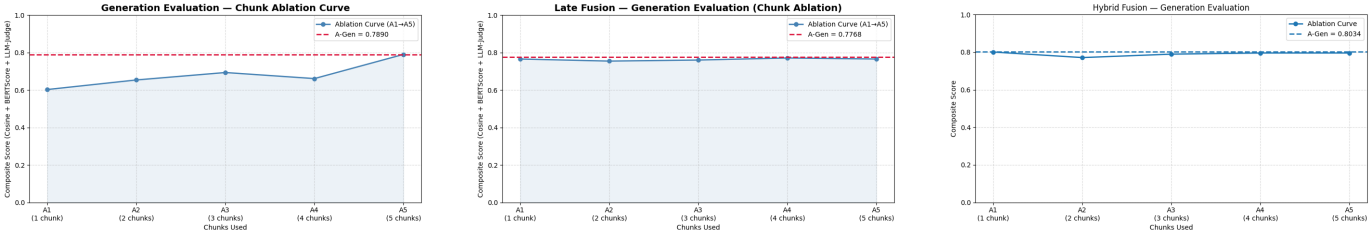


Fig. 4. Generation evaluation curves for the Early Fusion (left), Late Fusion (center), and Hybrid Fusion (right) pipelines.

TABLE II
GENERATION PERFORMANCE AND CHUNK-ABLATION BEHAVIOUR

Pipeline	A-Gen Score	Behaviour Across A1–A5	Interpretation
Early Fusion	0.7890	Scores generally increased as more chunks were provided, with a slight decrease at A4	Generation benefited considerably from additional retrieved context
Late Fusion	0.7768	Scores remained relatively stable across all chunk counts	Generation was robust and less sensitive to the number of retrieved chunks
Hybrid Fusion	0.8034	Scores remained consistently high, with only small variations	The generation component used the available context effectively

This suggests that the generation stage of Hybrid Fusion was stable and effectively used the available context. However, this strong generation performance should be considered alongside its weaker retrieval results. The result indicates that Hybrid Fusion generated good answers when useful context was available, but its retrieval component was less successful at consistently obtaining the correct source chunk [1].

C. Overall Comparison

Overall, Late Fusion provided the strongest retrieval performance because it achieved the highest MRR and the lowest Error Point. Early Fusion achieved the highest Hit@5 but was evaluated on a smaller set of questions. Hybrid Fusion produced the weakest retrieval scores but achieved the highest generation score.

These results show that retrieval and generation performance do not necessarily improve together [1], [2]. Late Fusion was more effective at identifying and ranking relevant chunks, whereas Hybrid Fusion appeared more effective at using the context once it had been retrieved [1], [4]. Therefore, the selection of a fusion strategy should consider both retrieval quality and the ability of the generation model to use the retrieved information [1], [11].

VI. CONCLUSION

This study compared three Multimodal Retrieval-Augmented Generation pipelines based on Early Fusion, Late Fusion, and Hybrid Fusion [8], [11], [14]. The results show that Late Fusion provided the best overall retrieval performance, with the highest MRR and the lowest Error Point, while Early Fusion achieved the highest Hit@5 score but was tested on fewer questions. The proposed Error Point metric and CWGF framework help evaluate both retrieval and answer generation more clearly. Overall, the findings show that the choice of fusion method affects the performance of an M-RAG system, and the most suitable pipeline may depend on the structure and modalities of the dataset [17], [18], [21]. Future work can evaluate these pipelines on larger and more diverse datasets [2], [11], [14].

VII. AI USE DECLARATION

We declare that this paper is our own independent work, with all core ideas, system design, experimentation, analysis, and conclusions developed by the authors. AI tools, such as OpenAI's ChatGPT, Anthropic's Claude, and xAI's Grok, were used only as supporting tools and were not responsible for generating the main intellectual contributions of this research. From an implementation perspective, these tools helped to code, debug, and optimize components of the M-RAG pipelines during development. From a research perspective, ChatGPT was used to explore relevant tools and techniques, including identifying Docling as a suitable document extraction tool for our use case. In Addition, AI tools were used to improve language, improve readability, and help with the restructuring and formatting of content and tables within the paper. All research ideas, experimental analysis, and final conclusions were developed through the authors' own collaborative efforts.

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